Education

Bachelor of Technology – Biotechnology
CGPA: 7.19
2022 – 2026

Senior Secondary Education (Class XII – CBSE)
Percentage: 96%
2021

Secondary Education (Class X – CBSE)
Percentage: 94%
2019

Experience

Technology Analyst Intern

June 2025 – Aug 2025

- Analyzed incident and performance data across **12 distributed systems** using SQL and Python (Pandas, NumPy).
- Identified multiple workflow issues that reduced recurring incidents and improved response prioritization.
- Built SQL-based datasets and **Power BI dashboards** to track system latency, incident frequency, and infrastructure health.
- Enabled faster operational reviews and improved infrastructure capacity planning.

Research Intern – Biomedical AI

Dec 2024 – Feb 2025

- Developed a **heartbeat detection system** using computer vision and deep learning.
- Used CNN, U-Net, OpenCV, and facial landmark tracking for remote photoplethysmography (rPPG) signal extraction.
- Applied Fourier Transform and signal processing techniques to isolate heartbeat signals from facial video streams.
- Trained models using TensorFlow and optimized them for real-time health monitoring applications.

Web3 Security Data Analytics Intern

June 2024 – Aug 2024

- Analyzed **200K+ blockchain transactions across multiple chains** to detect anomalies and security patterns.
- Improved anomaly detection accuracy by **15%** through structured analysis and dataset engineering.
- Built machine-learning-ready datasets for smart contract threat detection models.
- Designed taxonomy for over **1000 blockchain security incidents** to improve classification and detection pipelines.

Projects

Gene Expression Analysis for Cancer Biomarker Prediction

- Built a bioinformatics ML pipeline using **Python, scikit-learn, and Pandas** to classify cancer vs normal tissue samples.
- Processed **10,000+ gene expression samples** from biomedical datasets.
- Achieved **92% classification accuracy** using Random Forest with PCA-based feature selection.
- Identified **25 key biomarkers** differentiating cancer and normal tissues.
- Visualized gene clusters using Matplotlib and Seaborn.

Explainable AI System for Customer Churn Prediction

- Built a two-stage machine learning pipeline using **Logistic Regression and XGBoost** to predict customer churn.
- Implemented preprocessing, feature scaling, and stratified train-test evaluation.
- Integrated **LLM-based analysis** to generate natural language explanations for churn reasons from customer emails.
- Developed an interactive **Streamlit application** for real-time churn prediction and visualization.

Retail Profit Optimization Analysis

- Built a financial analytics pipeline using **Python (Pandas, financial data APIs) and Power BI**.
- Conducted segmentation and pricing diagnostics revealing **12% margin improvement opportunities**.
- Designed interactive BI dashboards with scenario simulations and executive insights.

Leadership & Achievements

Student Literary & Debating Society – Leadership Role

- Led a **team of 50 members** to organize large-scale events with **1000+ participants**.
- Managed event operations and budget planning for major academic competitions.
- Received recognition across multiple national and international debating competitions.

Skills

Programming
Python, C, Data Structures & Algorithms, Object-Oriented Programming

Machine Learning & AI
Deep Learning, Computer Vision, CNNs, U-Net, Feature Engineering, Model Optimization

Data Science & Tools
SQL, Pandas, NumPy, Statistics, TensorFlow, Scikit-learn, Power BI, Excel